

Physical TOCTOU Attacks Against Vision-Language-Action Robots

Exploiting Observation–Action Freshness in Embodied Control

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Draft v0.1 — concept and positioning only; no empirical results are claimed.

Abstract

Vision-Language-Action (VLA) policies increasingly execute temporally extended action chunks predicted from a finite set of visual and proprioceptive observations. This design improves throughput and motion consistency, but it also introduces a temporal trust assumption: state observed when an action is selected is assumed to remain sufficiently valid when that action is physically executed. Prior robotics work has documented the performance cost of stale observations, inference delay, and long open-loop execution horizons, while recent security work has exposed time-of-check to time-of-use (TOCTOU) vulnerabilities in LLM and GUI agents. We ask whether the same systems-security primitive becomes an exploitable physical attack surface in embodied VLA control.

We formulate *Physical TOCTOU* (P-TOCTOU): an adversary allows a robot to observe a legitimate scene, waits until the policy has committed to an action or action chunk, and then changes task-relevant physical state before execution completes. The attacker does not need to alter prompts, model weights, training data, robot software, middleware, or the observation already consumed by the policy. The core hypothesis is that an attacker can exploit the observation–execution gap itself, causing a robot to execute an otherwise legitimate action against a world state that no longer satisfies the action’s preconditions. We define observation–action age and the adversarial freshness window as candidate quantities for characterizing this risk, and outline a benchmark based on controlled post-observation scene changes across action horizons and attack timings. This draft develops the threat model and research framing only; empirical validation is intentionally left for subsequent work.

1 Introduction

Vision-Language-Action models are rapidly becoming a general interface between foundation models and physical

control. Modern VLA policies combine visual observations, language instructions, robot state, and learned action generation to produce low-level control trajectories. Many systems predict a sequence of future actions from a single policy query, commonly called *action chunking*. Action chunking amortizes expensive inference and improves temporal coherence, but necessarily creates intervals in which the robot continues to act without incorporating a new observation. Recent work has repeatedly identified the resulting freshness problem: long chunks reduce responsiveness to new information [7]; asynchronous execution introduces prediction–execution mismatch [5, 6]; and high-capacity VLA policies can make decisions from observations that are already stale by execution time [9]. These studies primarily treat stale state as a reliability, latency, or control problem.

Security research suggests a second interpretation. A state transition that occurs between validation and use is the classical structure of a time-of-check to time-of-use (TOCTOU) vulnerability. In software agents, *Mind the Gap* formalized TOCTOU in LLM-enabled workflows where files or API state can change after an agent checks them but before later use [1]. *Atomicity for Agents* independently studied the same class of vulnerability in browser-use agents [2]. For vision-driven computer-use agents, Xu formalized an observation-to-action gap in which an interface can be modified after a screenshot but before a click, describing the resulting failure as a visual atomicity violation [3]. These works demonstrate that agentic systems inherit security assumptions about state freshness even when their decision-making is learned rather than explicitly programmed.

We argue that embodied control creates a particularly consequential version of the same primitive. Consider a robot instructed to place a red object into a tray. At time t_o , the robot observes the object at pose x_o and a VLA policy predicts an action chunk $A = (a_1, \dots, a_H)$. After the observation has been consumed, but before all actions have executed, an external actor moves the object, tray, or obstacle configuration. The original observation was authentic, the policy output may have been locally correct, and no digital component needs to be compromised. Nevertheless, some suffix of A is now being applied to a

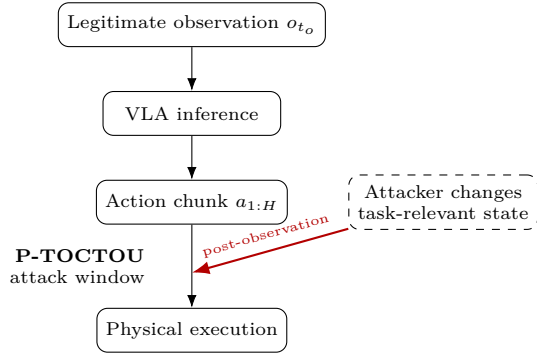


Figure 1: Conceptual Physical TOCTOU attack. The robot observes a legitimate state and commits to an action sequence. The attacker changes the physical world after observation but before the relevant action is used. No malicious visual input needs to be present at observation time.

physical state for which it was never validated.

This threat model is distinct from several nearby lines of work. Physical prompt injection attacks place adversarial text or visual content in the robot’s field of view so that malicious information is consumed during perception [11, 12]. Trajectory-level redirection attacks manipulate the language conditioning signal while keeping the environment fixed [13]. SilentDrift is especially close to our target because it explicitly weaponizes the open-loop interval created by action chunking; however, SilentDrift implants a training-time backdoor and relies on poisoned demonstrations to create stealthy within-chunk drift [4]. P-TOCTOU instead assumes an uncompromised policy and uncompromised training pipeline: the attacker acts only on the external physical state *after* a legitimate observation has already been accepted.

The closest non-adversarial VLA literature makes the proposed attack plausible but does not, by itself, establish it as a security phenomenon. REMAC identifies *intra-chunk inconsistency*, where parts of an action chunk become inconsistent with current perception during real-time execution [5]. DEFLECT treats latency as a source of prediction–execution mismatch and learns delay-robust behavior from fresh/stale counterfactual pairs [6]. Adaptive Action Chunking explicitly notes that longer chunks reduce responsiveness to new information [7]. A2C2 uses the latest observation to correct an existing action chunk at every control step [8], while recent reinforcement-learning work similarly separates slow VLA action generation from fast reactive edits conditioned on fresh state [9]. Act-Fovea is the closest runtime-safety system we found: it detects temporal inconsistency among visual observations, robot state, and actions, including visual delay and frozen-observation replay [10]. These methods address stale or disturbed state as a deployment failure. Our proposed contribution is to characterize *deliberately timed post-observation physical manipulation* as an adversarial capability with measurable attack windows.

We therefore define the central security question as follows: *Can an adversary who cannot influence the observation consumed by the policy, but can modify task-relevant physical state after that observation, reliably redirect or invalidate the physical outcome of an action-chunked VLA?* This formulation deliberately narrows the attacker’s capabilities. The adversary has no access to model parameters, training data, prompts, robot software, ROS middleware, network traffic, or policy outputs before execution. The adversary may only alter selected physical state after the policy’s relevant observation has been captured. This is stronger than ordinary environmental non-stationarity because the state change is chosen and timed to exploit the robot’s control cadence.

Two quantities may help turn this threat model into a measurable security property. First, we define the *observation–action age* of an executed action a_i as

$$\text{OAA}(a_i) = t_{\text{exec}}(a_i) - t_{\text{obs}}, \quad (1)$$

where t_{obs} is the time of the most recent task-relevant observation used to generate or validate a_i . OAA captures how old the supporting perception is when an action reaches the physical world. Second, we define an *adversarial freshness window* (AFW) as the interval after a relevant observation during which an attacker can alter physical state and still cause a target failure before the policy incorporates corrective evidence. Unlike raw inference latency, AFW is an end-to-end security quantity: it depends on sensing cadence, model latency, chunk horizon, controller behavior, task geometry, and any runtime revalidation mechanism.

The planned evaluation follows directly from this framing. A policy first observes an unmodified task and produces an action or action chunk. At controlled offsets after observation, the experiment changes one task-relevant physical variable—for example, object pose, receptacle pose, or obstacle state—without modifying the robot’s software inputs retrospectively. Experiments then vary action horizon and attack timing and measure whether the robot continues toward an outcome that is invalid under the updated world state. The key comparison is not merely attacked versus clean execution, but attack success as a function of OAA, chunk length, and the timing of state mutation. A lightweight defense can then revalidate task-relevant state immediately before committing irreversible or contact-sensitive actions, allowing the security benefit to be measured against added latency and false interruption cost.

This framing is intentionally conservative about novelty. Action chunking’s open-loop blind spot is established [4, 7]; stale observations and execution mismatch are established robotics problems [5, 6, 9]; TOCTOU is established for LLM, browser, and GUI agents [1, 2, 3]; and temporal VLA safeguarding is already an active research area [10, 14]. The intended contribution is narrower: to test whether these ingredients compose into a practical *physical, post-observation, inference-time attack* against

otherwise uncompromised VLA control, and to quantify the resulting freshness boundary as a security property rather than only a control-performance concern. Establishing that claim requires empirical evidence; this draft makes no claim that the hypothesis has yet been validated.

Planned contributions. Subject to empirical validation, a complete version of this work would aim to contribute: (i) a threat model for post-observation physical state manipulation against VLA policies; (ii) quantitative freshness metrics such as OAA and AFW; (iii) a controlled benchmark varying chunk horizon, attack timing, and scene mutation; and (iv) an evaluation of pre-execution state revalidation as a low-cost mitigation. The paper should avoid any “first” claim until a broader literature review and experimental replication have been completed.

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